

Alonso Marcos Muñoz

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[LinkedIn](#) | [Portfolio](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Data Engineer at Tragsatec, where I build and run Python and SQL ETL pipelines that integrate, validate and publish the data catalogue information of 22 Spanish ministries and public bodies. I cover the full cycle: modelling and transforming data in PostgreSQL and Oracle, automating quality checks and deploying to development, test and production. My MSc in Big Data and Cloud Computing extended this approach to modern platforms with Databricks, Spark, Kafka, Airflow and AWS.

PROFESSIONAL EXPERIENCE

Tragsatec

Albacete, Spain

Data Engineer

Oct. 2025 - Present

- Develop and maintain Python and SQL ETL pipelines that automatically collect, clean, validate and publish the data catalogue information of 22 Spanish ministries and public bodies.
- Design PostgreSQL and Oracle transformations, aggregations and views so that data arrives structured, consistent and ready to be queried and published.
- Automate quality checks that confirm every dataset meets the European open data standard before publication, so that errors do not reach official catalogues.
- Set up and run development, test, pre-production and production environments with Docker, Podman, Git and SSH, keeping deployments documented, repeatable and easy for the team to review.
- Within the AgoraData project, lead the development of the web interface that serves as the front end for all of ImpulsaDATA's backend and Python processes, using Java, Spring Boot, Vaadin and PostgreSQL. Coordinate and delegate security, testing, accessibility and CI/CD.

Tragsatec

Albacete, Spain

Data Engineer (Internship)

Jun. 2025 - Sep. 2025

- Designed and built the first version of the project's metadata conversion tool, a Python ETL process that maps source information to the European open data standard and publishes it in CKAN.

La Fábrica del Tiempo

Albacete, Spain

Data & Automation Engineer (Internship)

Feb. 2024 - Aug. 2024

- Automated lead management, notifications and approvals with Power Automate, connecting SharePoint, the CRM and Microsoft 365, improving the efficiency of the targeted processes by more than 25%.

Alcázar de San Juan City Council

Alcázar de San Juan, Spain

IT Technician (Internship)

Mar. 2019 - Jun. 2019

- Provided technical support to municipal staff and resolved hardware, software and network incidents in a public administration setting.

SELECTED APPLIED PROJECTS

- **Databricks Telco Churn Lakehouse:** co-developed end-to-end data platform that predicts customer churn for a telecom operator on synthetic data using PySpark, Delta Lake, Unity Catalog and MLflow:
<https://github.com/alonsomarcosm99/databricks-telco-churn-lakehouse>
- **Batch and streaming platform:** processes batch and real-time data with Spark, Kafka and Airflow and organises it into layers of increasing quality (Medallion architecture) on Delta Lake:
<https://github.com/alonsomarcosm99/spark-kafka-airflow-data-platform>
- **AWS Smart Parking:** cloud IoT platform that collects real-time data from 40 simulated parking sensors and serves it in a web dashboard, using AWS Lambda, DynamoDB and MQTT:
<https://github.com/alonsomarcosm99/smart-parking-albacete>

EDUCATION AND CERTIFICATIONS

UCLM - MSc Big Data and Cloud Computing

2025 - 2026

UCLM - BSc Computer Engineering, Information Technology

2019 - 2025

IES Juan Bosco - Higher Diploma in Networked Systems Administration

2017 - 2019

CAPM - Project Management Institute

B2 First - Cambridge English

TECHNICAL SKILLS

Data Engineering: Python, SQL, ETL, data modelling, PostgreSQL, Oracle, REST APIs, data quality and traceability.

Platforms and governance: CKAN, OpenMetadata, DCAT-AP-ES, RDF, JSON-LD, SHACL, Power Apps and Power Automate.

Applied Big Data and cloud: Databricks, PySpark, Spark, Delta Lake, Airflow, Kafka, MLflow, AWS and Terraform.

Development and operations: Java, Spring Boot, Vaadin, Docker, Podman, GitLab CI/CD, Git, Linux and SSH.

Languages: native Spanish and professional English (B2 First).